

Nov-18

today:

Partial
wave scattering

(6.4.4)

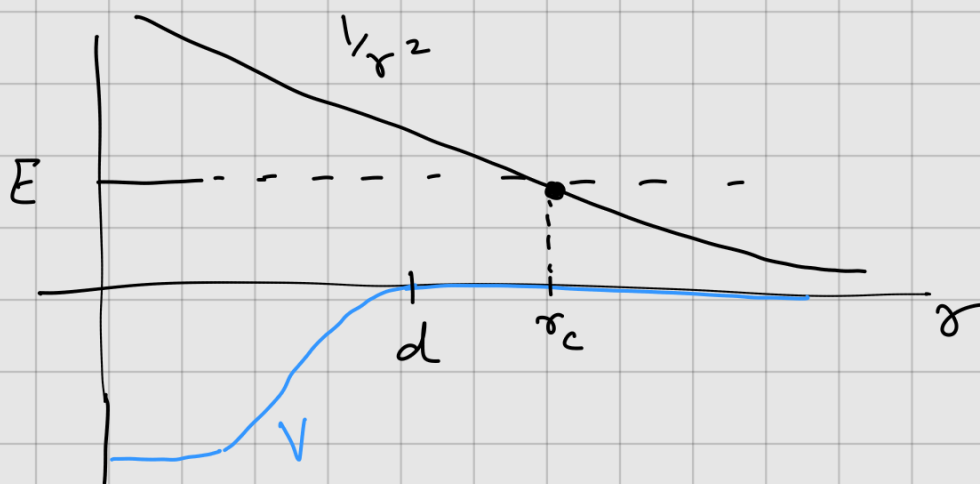
Zetilli

(11.7)

WARM-UP

$$\frac{-\hbar^2}{2m} u'' + \underbrace{\left(\frac{\hbar^2 l(l+1)}{2mr^2} + V(r) \right)}_{V_{\text{eff}}} u(r) = E u(r)$$

Suppose $V(r) = 0$ for $r > d$



(a) Find classical turning point r_c
 $V_{\text{eff}}(r_c) = E$; assume $r_c > d$

(b) Find l s.t. $r_c > d$
replace $E \rightarrow k$, $E = \frac{\hbar^2 k^2}{2m}$

$\rightarrow$ (a) $V_{\text{eff}} = E$

$$\frac{\hbar^2 l(l+1)}{2m r_c^2} + V(r_c) = E$$

↳ 0

$$\Rightarrow r_c = \left(\frac{\hbar^2 l(l+1)}{2m E} \right)^{1/2}$$

$$(b) \quad r_c = \left(\frac{\hbar^2 l(l+1)}{2m \frac{\hbar^2 k^2}{2m}} \right)^{1/2} > d$$

$$\Rightarrow \frac{l(l+1)}{k^2} > d^2$$

$$\Rightarrow l(l+1) > k^2 d^2$$

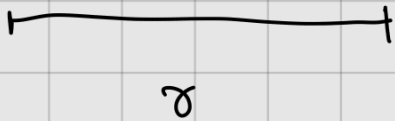
$$\Rightarrow \boxed{k d < \sqrt{l(l+1)}}$$

Demonstration:

$v \rightarrow$
chalk
projectile



$\uparrow$
Stickman
Dr. Sommer



$\rightarrow$
chalk would
keep moving without
interaction

$\Rightarrow$ the distance
b/w projectile
& obstacle keeps
on increasing.

which is exactly
what would happen
if there was interaction.

Plus the phase shift

* from Obstacle's perspective, interaction & no
interaction are almost identical processes
(at least in terms of evolution of relative dist.)

No scattering if $r_c > d$

$$kd \gtrsim 1$$

$\Rightarrow$ no scattering

$$\Rightarrow l^{\max} \approx kd$$

Partial Wave analysis

• Review: $V(\vec{r}) = V(r)$
 $\psi_{\text{inc}}(\vec{r}) = e^{ikz}$

Solve T.I.S.E

$$\psi(r, \theta) = \sum_{l=0}^{\infty} a_l R_l(r) P_l(\cos \theta)$$

substitute $u_l(r) = r R_l(r) \rightarrow$ Radial eqⁿ

$$\frac{-\hbar^2}{2m} u'' + \left[\frac{\hbar^2 l(l+1)}{2mr^2} + V(r) \right] u = \underbrace{\frac{\hbar^2 k^2}{2m}}_E u(r)$$

" using a result derived earlier using L.S.E & Green's fⁿ
 for large r

$$\psi(r, \theta) \rightarrow () \left[e^{ikz} + \frac{f(\theta) e^{ikr}}{r} \right]$$

$\hookrightarrow$ const. any $\equiv 1$

we know,

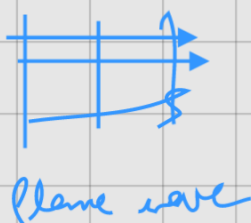
$$R_l(r) \xrightarrow{r \rightarrow \infty} \frac{1}{kr} \sin(kr - \frac{l\pi}{2} + \delta_l)$$

• solve for $R_l(r)$ or $u_l(r) \rightarrow \delta_l$
 use $\delta_l \rightarrow f$

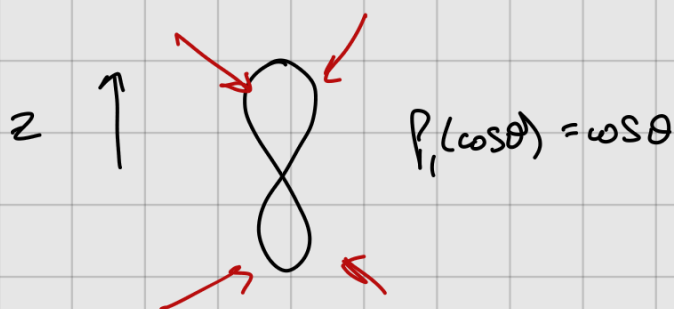
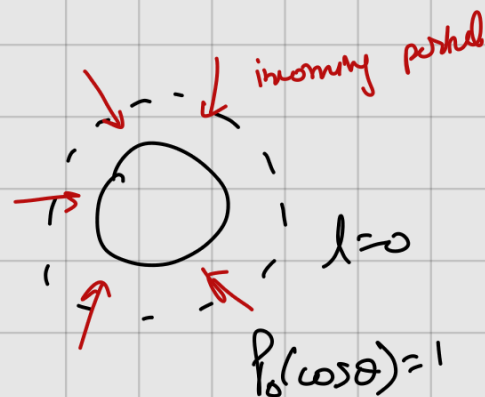
$$\frac{d\sigma}{d\Omega} = |f|^2$$

Result we want to derive

$$f(\theta) = \sum_{l=0}^{\infty} \frac{2l+1}{K} e^{i\delta_l} \sin(\delta_l) P_l(\cos\theta)$$



→
decompose
into spherical
waves



$$P_1(\cos\theta) = \cos\theta$$

the interaction with obstacle
introduces the phase, which
contains all the physics of
interaction.

doing the derivation:

$$\psi \rightarrow e^{ikz} + f(\theta) \frac{e^{ikr}}{r} \quad \text{as } r \rightarrow \infty$$

$$\psi(r, \theta) = \sum_{l=0}^{\infty} a_l \frac{1}{kr} \sin\left(kr - \frac{l\pi}{2} + \delta_l\right) P_l(\cos\theta)$$

$$= \sum_l \frac{a_l}{2ikr} \left(e^{i(kr - \frac{l\pi}{2} + \delta_l)} - e^{i(kr - \frac{l\pi}{2} + i\delta_l)} \right) P_l$$

$$= \sum_l \left(\frac{a_l}{2ikr} (-i)^l e^{i(kr + i\delta_l)} - \frac{a_l}{2ikr} i^l e^{-i(kr + \delta_l)} \right) P_l$$

expand e^{ikz}

$$e^{ikz} = \sum_{l=0}^{\infty} (2l+1) i^l j_l(kr) P_l(\cos\theta)$$

$$\rightarrow \sum_l \frac{(2l+1) i^l}{2ikr} [(-i)^l e^{i(kr)} - i^l e^{-i(kr)}] P_l(\cos\theta)$$

equating:

$$\psi - e^{ikz} \rightarrow \sum_l \left\{ \frac{e^{i(kr)}}{2ikr} (-i)^l [e^{i\delta_l} a_l - (2l+1)i^l] - \frac{e^{-i(kr)}}{2ikr} i^l [a_l e^{-i\delta_l} - (2l+1)i^l] \right\} P_l(\cos\theta)$$

goes to zero.

$$a_l = (2l+1) i^l e^{i\delta_l}$$

$$f(0) = \sum_{l=0}^{\infty} \frac{(-i)^l}{2ik} [(2l+1) i^l e^{2i\delta_l} - (2l+1) i^l] P_l(\cos\theta)$$

$$= \sum_{l=0}^{\infty} \frac{2l+1}{2ik} [e^{2i\delta_l} - 1] P_l(\cos\theta)$$

$$= \sum_{l=0}^{\infty} \frac{(2l+1)}{k} e^{i\delta_l} \sin(\delta_l) P_l(\cos\theta)$$

total cross section

$$\sigma = \int |f|^2 d\Omega$$

$$= \int d\Omega \sum_{l, l'} \frac{(2l+1)(2l'+1)}{k^2} e^{i\delta_l} e^{-i\delta_{l'}} \sin \delta_l \sin \delta_{l'} P_l(\cos\theta) P_{l'}(\cos\theta)$$

↓
dummy variable

use: $\int_0^\pi P_l(\cos\theta) P_{l'}(\cos\theta) \sin\theta d\theta = \frac{2}{2l+1} \delta_{ll'}$

$$\sigma = \sum_l \frac{(2l+1)^2}{k^2} \sin^2 \delta_l \left(\frac{4\pi}{2l+1} \right)$$

$$\sigma = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l$$